Earned Compression: Categorical Decomposition of Petri Nets via Tropical Analysis and ZK Observation

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Abstract

We prove that the *core-observer decomposition* of Petri nets is invariant across three independent formalisms: ODE simulation, tropical spectral analysis, and zero-knowledge proof. Given a Petri net N with incidence matrix $C \in \mathbb{Z}^{|T|\times|P|}$, each formalism independently discovers the same structural boundary—a partition $T = T_{\text{core}} \sqcup T_{\text{obs}}$ where the core subnet admits the formalism's compression and the observer subnet does not:

- **ODE simulation:** the core evolves continuously; the observer is evaluated at equilibrium.
- **Tropical analysis:** the core is a timed event graph with max-plus eigenvalue λ; the observer's places have multiple consumers, breaking linearity.
- **ZK encoding:** the core compiles to R1CS arithmetic; the observer requires a separate witness structure.

The convergence of three formalisms on the same decomposition is itself the proof that the boundary is structural, not an artifact of any particular analysis. We formalize this as *earned compression*: each formalism applies a structure-preserving map that discards exactly the information irrelevant to its target property, and the earned qualifier means each compression is proved to preserve the target property exactly (no lossy approximation).

We validate on tic-tac-toe, where the core net (18 play transitions, 33 places) is an event graph with eigenvalue $\lambda = 1$, and the observer net (17 win/draw transitions) is a pure-sink layer that consumes cell tokens to detect victory. We extend the analysis to payment settlement networks, defining the free symmetric monoidal category **Settle** whose string diagram is both the protocol specification and the throughput certificate. The same incidence matrix C serves all three analyses without modification.

The unifying principle is Kolmogorov-flavored: a domain is *understood* when its shortest description is also its most useful description. The incidence matrix is that description for Petri-net-structured domains.

1 Introduction

The central claim of this paper is:

> *The core-observer decomposition of Petri nets is invariant across simulation, proof, and analysis formalisms. Each formalism independently discovers the same structural boundary. The convergence is the proof.*

The central object is the incidence matrix $C \in \mathbb{Z}^{|T|\times|P|}$ of a Petri net. We show that C is a *maximally compressed* representation: the shortest description from which all behavioral, structural, and cryptographic properties can be recovered without loss.

Three formalisms, developed independently, each discover the same partition of the net's transitions into a compressible *core* and an incompressible *observer*:

1. **ODE simulation** [15]. The mass-action ODE system $\dot{\mathbf{x}} = C \cdot \mathbf{v}(\mathbf{x})$ evolves the core continuously to equilibrium $x_i^* = 1/n_i$. The observer is evaluated at the fixed point—it does not participate in the continuous dynamics.

2. **Tropical analysis** (this paper). The max-plus eigenvalue λ of the tropical adjacency matrix derived from C_{core} compresses the entire reachability dynamics to a throughput scalar. The observer subnet fails the timed event graph property (multiple consumers per place) and is *inexpressible* in the tropical semiring.

3. **ZK encoding** (this paper and [15]). The incidence matrix compiles directly into R1CS constraints. The core transitions become arithmetic constraints ($\mathbf{m}' = \mathbf{m} + \Delta_t$); the observer requires a separate witness structure because its consumption pattern breaks the uniform delta structure.

The three compressions are compatible: they operate on the same matrix C, discover the same boundary, and preserve disjoint properties. No single formalism could prove the boundary is structural; the convergence of three independent formalisms does.

Running example. We use tic-tac-toe throughout. The core net has 33 places and 18 transitions (9 cells $\times$ 2 players, with turn control). The observer adds 17 transitions (8 win lines $\times$ 2 players + 1 draw) that consume cell tokens and the `game_active` flag. The full net has 33 places, 35 transitions, and 189 arcs.

Contributions.

- The *formalism-invariance theorem*: three independent analyses discover the same core-observer boundary, proving the decomposition is structural (Section 3).
- A formal definition of the core-observer decomposition, with the categorical characterization as a functor $F : \mathbf{Petri} \to \mathbf{Set}$ (Section 4).
- A proof that the core subnet of TTT is a timed event graph with tropical eigenvalue $\lambda = 1$ (Section 5).
- A proof that the observer subnet is *not* tropical, and that this is the minimal necessary complexity for win detection (Section 7).
- The earned compression pipeline: each stage is a structure-preserving map that discards exactly the irrelevant information (Section 6).
- Experimental validation via the `pflow-tropical` Rust crate, all reproducible from C alone (Section 8).
- The category **Settle**: a free symmetric monoidal category for payment networks, where the string diagram is the protocol specification, the tropical eigenvalue is the throughput certificate, and conservation is an accounting theorem (Section 9).

2 Background

2.1 Petri Nets

A *Petri net* is a bipartite directed graph $N = (P, T, A^-, A^+)$ where P is a finite set of *places*, T is a finite set of *transitions*, and $A^-, A^+ : T \times P \to \mathbb{N}$ are the *input* and *output* incidence functions. A *marking* $\mathbf{m} \in \mathbb{N}^{|P|}$ assigns a non-negative integer token count to each place. Transition t is *enabled* at marking $\mathbf{m}$ if $\mathbf{m}(p) \geq A^-(t,p)$ for all $p \in P$. Firing t produces the successor marking $\mathbf{m}' = \mathbf{m} + \Delta_t$, where the *delta vector* $\Delta_t(p) = A^+(t,p) - A^-(t,p)$.

The *incidence matrix* $C \in \mathbb{Z}^{|T| \times |P|}$ has rows $C_t = \Delta_t$.

2.2 Tropical Semiring

The *tropical semiring* (max-plus algebra) is the set $\mathbb{R}_{\max} = \mathbb{R} \cup \{-\infty\}$ equipped with:

$$
\begin{aligned}
a \oplus b &= \max(a,b) && \text{(tropical addition)} \\
a \otimes b &= a + b && \text{(tropical multiplication)}
\end{aligned}
$$

with identity elements $\mathbb{0} = -\infty$ and $\mathbb{1} = 0$.

For a square matrix $A \in \mathbb{R}_{\max}^{n \times n}$, the *tropical eigenvalue* $\lambda(A)$ is the maximum circuit mean:

$$\lambda(A) = \max_{\gamma} \frac{w(\gamma)}{|\gamma|}$$

where the maximum ranges over all directed circuits γ in the precedence graph of A, $w(\gamma)$ is the total weight of the circuit, and $|\gamma|$ is its length.

2.3 Timed Event Graphs

A *timed event graph* (TEG) is a Petri net where every place has exactly one input transition and exactly one output transition. TEGs are the subclass of Petri nets whose dynamics are exactly captured by tropical linear algebra: the evolution of firing times satisfies

$$\mathbf{x}(k+1) = A \otimes \mathbf{x}(k)$$

where A is the tropical adjacency matrix and $\mathbf{x}(k)$ is the vector of k-th firing times [1].

The asymptotic throughput of a TEG equals the tropical eigenvalue $\lambda(A)$: each transition fires once per λ time units in steady state.

2.4 Zero-Knowledge Proofs

A *zero-knowledge succinct non-interactive argument of knowledge* (zk-SNARK) allows a prover to convince a verifier that a statement is true without revealing the witness. Groth16 [8] produces proofs of constant size (128 bytes over BN254) with verification in $O(1)$ pairings. The statement is encoded as a *rank-1 constraint system* (R1CS): a system of equations $Az \circ Bz = Cz$ where $\circ$ is the Hadamard product.

For Petri net transitions, the R1CS encodes: given public pre-state root $h(\mathbf{m})$ and post-state root $h(\mathbf{m}')$, there exists a transition index t such that $\mathbf{m}' = \mathbf{m} + \Delta_t$ and $\mathbf{m}(p) \geq A^-(t,p)$ for all p.

3 Formalism Invariance

The key result is not any single compression, but that three independently motivated formalisms converge on the same structural boundary.

Theorem 1 (Formalism Invariance of Core-Observer Decomposition)**.** *Let $N = (P, T, A^-, A^+)$ be a Petri net with incidence matrix C. If the partition $T = T_{\text{core}} \sqcup T_{\text{obs}}$ satisfies:*

1. ***ODE:*** *the mass-action system $\dot{\mathbf{x}} = C_{\text{core}} \cdot \mathbf{v}(\mathbf{x})$ admits a stable equilibrium, while C_{obs} contributes only sink terms evaluated at that equilibrium;*

2. ***Tropical:*** *N_{core} is a timed event graph (each place has one input and one output transition), while N_{obs} violates this property (places shared with the core have multiple consumers);*

3. ***ZK:*** *C_{core} compiles to uniform R1CS constraints ($\mathbf{m}' = \mathbf{m} + \Delta_t$ with boolean transition selectors), while C_{obs} requires auxiliary witness variables for its non-uniform consumption patterns;*

then the three partitions agree: the same transitions are classified as core (resp. observer) by all three formalisms.

Proof sketch. The three conditions are entailed by a single structural property: the *sink property* of Definition 3. A transition t is in T_{obs} if and only if it consumes from core places without producing into them.

ODE direction: A sink transition has $A^+(t,p) = 0$ for all core places p, so its contribution to $\dot{\mathbf{x}}$ is purely dissipative. The core reaches equilibrium independently; the observer evaluates the result.

Tropical direction: A sink transition adds a consumer to each core place it reads, breaking the one-consumer constraint. Conversely, any transition that preserves the event-graph property must produce into every place it consumes from—which is the definition of a core transition.

ZK direction: Core transitions have uniform delta structure ($\mathbf{m}' = \mathbf{m} + \Delta_t$), where the multiplexer selects Δ_t from the incidence matrix columns. Observer transitions break uniformity: they consume from places whose tokens are shared with other observer transitions, requiring auxiliary witness variables to prove non-interference.

Since all three classifications reduce to the same structural criterion (sink property), the partitions agree. □

Remark 2. *No single formalism can prove the boundary is structural: the ODE analysis might be an artifact of mass-action kinetics; the tropical analysis might be an artifact of event-graph restrictions; the ZK encoding might be an artifact of R1CS structure. But the convergence of three independent formalisms—each with different mathematical foundations, different computational models, and different target properties—on the* same *boundary constitutes evidence that the boundary is intrinsic to the net, not to the analysis.*

4 The Core-Observer Decomposition

Definition 3 (Core-Observer Decomposition)**.** *Let $N = (P, T, A^-, A^+)$ be a Petri net. A* core-observer decomposition *is a partition $T = T_{\text{core}} \sqcup T_{\text{obs}}$ such that:*

1. *The* core subnet *$N_{\text{core}} = (P, T_{\text{core}}, A^-|_{T_{\text{core}}}, A^+|_{T_{\text{core}}})$ is the restriction of N to core transitions.*

2. *The* observer subnet *$N_{\text{obs}} = (P, T_{\text{obs}}, A^-|_{T_{\text{obs}}}, A^+|_{T_{\text{obs}}})$ satisfies the* sink property*: for every place p shared with the core, $A^+(t,p) = 0$ for all $t \in T_{\text{obs}}$. That is, observer transitions consume from core places but never produce into them.*

The sink property ensures that the observer cannot influence the core's dynamics: if no observer transition fires, the core evolves identically whether or not the observer exists. The observer is a *passive monitor* that reads state by consuming tokens, and its firing terminates the system (or a subsystem).

Proposition 4 (Observer Preserves Core Reachability)**.** *Let $N = N_{\text{core}} + N_{\text{obs}}$ be a core-observer decomposition. If $\sigma = t_1 t_2 \cdots t_k$ is a firing sequence of N_{core} from marking $\mathbf{m}_0$, then σ is also a firing sequence of N from $\mathbf{m}_0$, and the resulting markings on all places agree.*

Proof. By the sink property, observer transitions do not produce into core places, so they cannot increase the token count on any place that a core transition consumes. Since no observer transition fires in σ, the enabledness conditions for each $t_i \in T_{\text{core}}$ are identical in N and N_{core}. $\square$

TTT example. The tic-tac-toe net decomposes as:

- T_{core}: 18 play transitions (`x_play_00`, ..., `o_play_22`).
- T_{obs}: 16 win transitions (`x_win_row0`, ..., `o_win_anti`) + 1 draw transition.

Each win transition consumes 3 cell tokens (x_{ij} or o_{ij}), the `game_active` flag, and a turn token; it produces a single `win_x` or `win_o` token. The cell tokens are not reproduced—the game is over.

5 Tropical Analysis of the Core

5.1 Constructing the Tropical Adjacency Matrix

Given the core subnet N_{core} with incidence matrix C_{core}, we construct the tropical adjacency matrix $A \in \mathbb{R}_{\max}^{|T_{\text{core}}|\times|T_{\text{core}}|}$ as follows:

For transitions $t_i, t_j \in T_{\text{core}}$, set

$$A[t_i, t_j] = \max_{p \in P}\{A^-(t_i, p) : A^+(t_j, p) > 0 \text{ and } A^-(t_i, p) > 0\}$$

with $A[t_i, t_j] = -\infty$ if no such place exists.

That is, $A[t_i, t_j]$ is the weight of the heaviest arc connecting an output of t_j to an input of t_i through a shared place.

5.2 TTT Core is Tropical

Theorem 5 (TTT Core Eigenvalue)**.** *The tropical adjacency matrix of the TTT core subnet has eigenvalue $\lambda = 1$.*

Proof sketch. The core net's turn-control places (`x_turn`, `o_turn`) create a bipartite structure: every X-play transition produces into `o_turn`, and every O-play transition produces into `x_turn`. The shortest circuit is any X-play $\to$ O-play pair (length 2, total weight 2), giving $\lambda = 2/2 = 1$. All longer circuits (via `move_tokens`) have the same mean weight. $\square$

Remark 6 (Structural vs. Behavioral Tropicality)**.** *The core's claim to tropical status requires qualification. A timed event graph requires each place to have* exactly one *input transition and* exactly one *output transition. The cell places (`p00`, `x00`, `o00`, etc.) satisfy this: each is produced by one transition and consumed by one transition. The turn-control places do not: `x_turn` has 9 producers (all `o_play` transitions) and 9 consumers (all `x_play` transitions).*

Structurally, the turn places violate the TEG property. Behaviorally, they do not: a single token occupies each turn place, and mutual exclusion via the cell places ensures exactly one transition fires per turn. The token ping-pongs 1-to-1 through reachable markings, even though the structure *has 9-to-9 fan.*

This gives a three-layer decomposition with decreasing tropicality:

Layer	*Places*	*Structural TEG*	*Behavioral TEG*
Cell (core)	*pij, xij, oij*	✓	✓
Turn control	*x_turn, o_turn*	× *(9:9 fan)*	✓ *(mutex)*
Observer	*win_x, win_o*	×	×

The standard Petri net subclass hierarchy sharpens the distinction. A free choice net *[4] requires that choice and synchronization never overlap: if a place p has arcs to transitions t_1 and t_2, then p must be the only input place of both.*

Net variant	***Petri net class***	***Why***
Core without turns	*State machine*	*Each transition has 1 input, 1 output place*
Core with turns	*Not free choice*	*Choice ($p_{ij} \to X$ or O) overlaps sync (+ turn token)*
Full (with observer)	*General*	*Win transitions synchronize on 3 cell places*

Without turn places, the core is 9 independent state machines (each cell has states $\{p, x, o\}$ with transitions {`x_play`, `o_play`})—a strict subclass of free choice nets. Adding turn enforcement breaks free choice: `x_play_11` requires both `p11` (cell empty) and `x_turn` (correct turn), so choice and synchronization overlap at `p11`.

This means the turn-enforced core sits in a gap in the standard taxonomy. It is not a timed event graph (turn places have 9:9 fan), not a state machine (transitions synchronize on two inputs), and not a free choice net (choice/sync overlap). Yet behaviorally it acts like an event graph: the turn token ping-pongs deterministically, and mutual exclusion ensures at most one transition fires per turn.

The three formalisms of Section 3 each see the turn places differently, exposing a divergence between structural and reachability-based analysis:

- ***ODE (continuous):*** *The turn token is a real-valued concentration. Mass-action kinetics naturally enforces mutual exclusion: when any `x_play` fires, the `x_turn` concentration depletes, suppressing the others. The ODE does not distinguish 9:9 fan from 1:1—the dynamics are smooth and the system is well-behaved.*
- ***Tropical (structural):*** *The tropical adjacency matrix sees all 9 producers and 9 consumers on `x_turn` and reports causal paths between transitions that can never actually co-fire. The structural analysis* overapproximates*—it cannot see the mutex that the cell places impose. This is precisely where the TEG property breaks.*
- ***Discrete (firing/RNN):*** *The Petri net firing rule enforces mutual exclusion exactly: one token, one consumer fires. An RNN trained on discrete trajectories from the turn-enforced core never observes non-alternating play—the net prevents it. The same RNN on the core* without *turn places learns to "cheat" by firing the same player's transitions consecutively, exploiting the structural gap.*

The turn places are the exact boundary where continuous and discrete analysis agree (mutual exclusion holds) but structural analysis diverges (it cannot verify mutual exclusion from the incidence matrix alone). This is a stronger statement than "behavioral TEG": it shows that the incidence matrix C—which all three formalisms share as input—is insufficient *to classify tropicality at the turn-control layer. Reachability information is required, and reachability is inherently a discrete state-space concept that the tropical semiring cannot express.*

Whether there exists a useful formal weakening—"tropical under mutual exclusion," or following the Petri net convention of [4], perhaps "weakly tropical"—that captures this behavioral event-graph property without requiring full state-space enumeration is an open question.

5.3 Why the Observer is Not Tropical

The observer subnet fails the event-graph property because cell places have multiple consumers. Consider place `x11` (X's mark on the center cell):

- **Producer:** `x_play_11` (exactly one).

- **Consumers:** `x_win_row1`, `x_win_col1`, `x_win_diag`, `x_win_anti` (four).

With four consumers, `x11` violates the TEG requirement of exactly one output transition per place. This is not an artifact of the encoding—it is the *irreducible complexity* of win detection: cell $(1,1)$ participates in four win lines, and each must independently check for three-in-a-row.

Proposition 7 (Observer Complexity is Tight)**.** *Any Petri net encoding of $n \times n$ tic-tac-toe win detection requires at least $2n+2$ transitions (for $2n+2$ win lines) and at least n^2 places with fan-out > 1 (for cells on the diagonal).*

6 The Compression Pipeline

We now formalize the unifying principle. Each analysis of the incidence matrix C is a *compression*: a structure-preserving map that discards information irrelevant to a target property while preserving the property exactly.

Definition 8 (Earned Compression)**.** *Let S be a structured object and $\pi(S)$ a property of S. A map $\kappa : S \to S'$ is an* earned compression *with respect to π if:*

1. *$|S'| < |S|$ (the output is strictly smaller),*
2. *$\pi(S) = \pi'(S')$ (the property is recoverable from the compressed form),*
3. *no map $\kappa' : S \to S''$ with $|S''| < |S'|$ satisfies condition 2 (the compression is tight).*

The tightness condition (3) is what distinguishes earned compression from approximation. It is the Kolmogorov complexity claim: S' is the shortest description of S from which π can be recovered.

6.1 The Four Stages

Stage	**Input**	**Output**	ρ	**Property preserved**
Domain $\to$ Net	Game rules	$C \in \mathbb{Z}^{\lvert T\rvert \times \lvert P\rvert}$	—	All firing semantics
Net $\to$ Eigenvalue	C (35×33)	$\lambda \in \mathbb{R}_{\max}$	1155	Throughput (core)
Net $\to$ Invariants	C	$\ker(C), \ker(C^T)$	$\lvert P\rvert / \dim \ker$	Conservation laws
Net $\to$ Observation	$\mathbf{m}, t_{\text{obs}}$	verdict $\in \{w, l, d\}$	5–10	Game outcome
Net $\to$ ZK proof	$C, \mathbf{m}, t$	$\pi \in \{0,1\}^{128}$	$\sim 10^4$	Transition validity

Each row is an earned compression: the output is strictly smaller than the input, the target property is exactly recoverable, and no smaller output suffices. The compression ratio ρ quantifies what is earned at each stage—the ratio of input size to output size, measuring exactly how much irrelevant information is discarded. Observation (Section 7) is the only stage that is *irreversible*: the consumed tokens are destroyed, not merely summarized.

Stage 1: Domain to net. The incidence matrix C is the compression of the game rules. Every legal state transition is encoded as a column of C; every conservation law is in $\ker(C)$; every reproducible firing sequence is in $\ker(C^T)$. The claim that C is the *shortest* such encoding is the core claim of the Petri net formalism: adding a place or transition that doesn't change the reachability graph is redundant; removing one that does changes the semantics.

Stage 2: Net to eigenvalue. For a timed event graph, the tropical eigenvalue λ determines the asymptotic firing rate of every transition. This compresses the full reachability graph (which may be exponential in the number of places) to a single scalar. The proof that λ is tight is classical [1]: λ is the unique value such that $A \otimes \mathbf{v} = \lambda \otimes \mathbf{v}$ has a solution, and no coarser summary determines the throughput.

Stage 3: Net to invariants. P-invariants ($\ker(C)$) are the conservation laws: weighted sums of token counts that remain constant under all firings. T-invariants ($\ker(C^T)$) are the reproducible firing sequences: multisets of transitions that return the net to its original marking. Both are computed from C alone and are topological—they depend on connectivity, not on weights or initial marking.

Stage 4: Net to ZK proof. The Groth16 proof compresses the statement "transition t is valid from marking $\mathbf{m}$ to marking $\mathbf{m}'$" into a 128-byte proof with $O(1)$ verification. The incidence matrix C is baked into the circuit as constants; the proof witnesses the private marking and transition index. This is the ultimate compression: the verifier learns *only* that the transition was valid, and nothing else about the state.

7 Observation as Irreversible Compression

7.1 Observation = Irreversibility

The core-observer decomposition is not merely structural—it is *thermodynamic*. The core subnet is reversible in the invariant-theoretic sense: P-invariants guarantee that every token consumed by a core transition is produced elsewhere. The total weighted token count is conserved across all core firings. No information is destroyed.

The observer subnet is irreversible. A win transition consumes cell tokens and the `game_active` flag, producing a single verdict token (`win_x`, `win_o`). The board state—which cells were marked, in what order—is destroyed. Only the verdict survives.

Definition 9 (Observation)**.** *An* observation *on a Petri net N is a transition $t \in T_{\text{obs}}$ such that:*

1. *t consumes from places shared with T_{core} (it reads the core's state),*
2. *t does not produce into any place shared with T_{core} (the read is destructive),*
3. *t produces into places exclusive to T_{obs} (it emits a verdict).*

An observation is irreversible*: no firing sequence from the post-observation marking can restore the pre-observation marking.*

This is not an analogy to measurement in physics—it is the same structure. A measurement apparatus couples irreversibly to a reversible system, producing a classical outcome that cannot be undone. The formalization here makes precise what "irreversible" means: the consumed tokens are not reproduced, so the pre-observation marking leaves the reachability set.

Proposition 10 (Observation requires irreversibility)**.** *Let $t \in T_{\text{obs}}$ be an observer transition. If t were reversible—i.e., if there existed a transition t^{-1} such that firing t then t^{-1} restores the original marking—then t would not terminate the system. The game could be "un-won." Irreversibility is not overhead; it is the definition of having observed.*

7.2 Quantifying the Earned Compression

Each observation has a measurable *compression ratio*: the information destroyed relative to the verdict produced.

Definition 11 (Observation Compression Ratio)**.** *For an observer transition t, let* $\text{consumed}(t) = \sum_p A^-(t,p)$ *be the total tokens consumed and* $\text{produced}(t) = \sum_p A^+(t,p)$ *be the total tokens produced. The* compression ratio *is:*

$$\rho(t) = \frac{\text{consumed}(t)}{\text{produced}(t)}$$

For TTT win transitions: each consumes 5 tokens (3 cell tokens + `game_active` + turn token) and produces 1 token (`win_x` or `win_o`), giving $\rho = 5$. The draw transition consumes 10 tokens (9 `move_tokens` + `game_active`) and produces 1, giving $\rho = 10$.

The compression ratio quantifies what is *earned*: 5 tokens of board state are irreversibly consumed to produce 1 bit of verdict. The 4 tokens of information destroyed are exactly the tokens that are irrelevant to the verdict—the specific cells that formed the winning line don't matter once you know who won.

Observer transition	**Consumed**	**Produced**	ρ	**Information discarded**
`x_win_*` (8 transitions)	5	1	5	Which 3 cells formed the line
`o_win_*` (8 transitions)	5	1	5	Which 3 cells formed the line
`draw`	10	1	10	Full move history (9 moves)
Core transitions (18)	3	3	1	None (reversible)

The core transitions have $\rho = 1$: every token consumed is produced elsewhere. This is the formal signature of reversibility. Observer transitions have $\rho > 1$: more is consumed than produced, and the difference is the earned compression—the exact amount of state that the verdict makes irrelevant.

7.3 Categorical Interpretation

The observer defines a functor $F : \mathbf{Petri} \to \mathbf{Set}$ that maps each net N to its set of terminal states (win, loss, draw) and each simulation morphism to the induced map on terminal states.

Proposition 12. *The observer functor F preserves the core's reachability: for any marking $\mathbf{m}$ reachable in N_{core}, the set of terminal states reachable from $\mathbf{m}$ in N is determined by the observer transitions enabled at $\mathbf{m}$.*

The functor F is the categorical formalization of the compression: it maps the exponentially large reachability graph to a finite set of verdicts. The compression ratio ρ measures the per-transition cost of applying F—the tokens of state traded for one token of verdict. The sink property (Definition 3) ensures F is well-defined: observer transitions cannot create new core-reachable markings, so observation is a one-way projection from dynamics to outcome.

8 Experiments

All experiments use the `pflow-tropical` Rust crate, which implements the `NetMatrix` type—a generic Petri net representation independent of the pflow ecosystem. The crate computes tropical adjacency matrices, eigenvalues, P/T-invariants, and discrete firing from the incidence matrix alone.

8.1 TTT Core: Tropical Eigenvalue

The core subnet (18 transitions, 33 places) has tropical eigenvalue $\lambda = 1.0$, confirming Theorem 5. The critical circuit has length 2 (one X-play, one O-play), corresponding to the alternating turn structure.

The tropical adjacency matrix correctly predicts transition dependencies:

- `x_play_00` enables `o_play_*` via `o_turn` (finite entry).
- `x_play_00` does not self-enable (entry $= -\infty$).
- `x_play_00` enables `x_win_row0` via `x00` (cross-layer dependency).

8.2 TTT Observer: Non-Tropical Verification

The full net (35 transitions) has places with fan-out > 1:

- `x11`: consumed by `x_win_row1`, `x_win_col1`, `x_win_diag`, `x_win_anti`.
- `game_active`: consumed by all 17 observer transitions.

The tropical adjacency matrix of the full net contains entries that don't correspond to true causal dependencies (e.g., win transitions appear to enable play transitions through shared control places), confirming that tropical analysis is not applicable to the observer layer.

8.3 Invariants

The core subnet has 9 P-invariants of the form $p_{ij} + x_{ij} + o_{ij} = 1$ (cell conservation: each cell is exactly one of empty, X, or O) plus turn and move-count invariants. These are purely topological—they hold for any arc weight, confirming the "structure trumps weights" principle.

8.4 RNN Self-Play and Hidden-State Sufficiency

An Elman RNN trained on the Petri net's marking vector learns to play $N \times N$ tic-tac-toe via self-play, with no game-specific architecture. The input is the raw marking $\mathbf{m} \in \mathbb{Z}^{|P|}$, the output is a distribution over transitions, and legal moves are enforced by masking disabled transitions to $-\infty$ before softmax. The same code runs unchanged on 3×3 (29 places, 18 transitions), 4×4 (50 places, 32 transitions), and 5×5 (77 places, 50 transitions) nets produced by the parameterized generator `builder_ttt_nxn(n)`.

Symmetric self-play convergence. Using per-move REINFORCE with symmetric rewards (each player receives +1 for winning, −1 for losing, +0.1 for drawing, from its own perspective), 3×3 converges to 96% draws—approaching optimal play, which is always a draw under perfect play from both sides. Larger boards (4×4, 5×5) show slower convergence but learn well below the random baseline, with the same architecture and hyperparameters.

Hidden-state sufficiency. A swapped-tuple bug provided an accidental ablation: in the 4×4 experiment, the RNN's `step()` function returns $(\mathbf{h}_t, \mathbf{y}_t)$ (hidden state, logits), but the game-play code destructured them as $(\mathbf{y}_t, \mathbf{h}_t)$—using the hidden state *as* the logit vector over transitions.

For 4×4, hidden size equals the number of transitions (= 32), so no dimension mismatch occurred. The RNN played coherent games with 73–100% confidence on many moves, executed recognizable strategies (column wins), and completed 10 rounds of self-play—all using $\tanh(W_{xh}\mathbf{x} + W_{hh}\mathbf{h} + \mathbf{b})$ directly as move probabilities.

For 5×5, hidden size (48) < transitions (50), producing an index-out-of-bounds panic that revealed the bug.

Remark 13 (Hidden State as Implicit Policy)**.** *The fact that the hidden state produces coherent play without the output projection W_{hy} implies that the RNN's internal representation is already organized around the transition space of the Petri net. Each hidden dimension tracks information relevant to* whether a specific transition should fire*, not generic sequential features. The output layer W_{hy} acts as a linear remix of information that is already present—it improves calibration but is not structurally necessary for action selection.*

This echoes a known phenomenon in representation learning: penultimate layers often encode richer task-relevant structure than the final output [14]. Here the "task" is the Petri net's firing semantics, and the hidden state's alignment with the transition space arises without any architectural bias toward it—the net's topology induces the structure through training alone.

Scaling. Table 1 shows that net size grows as $\Theta(n^2)$ while RNN parameter count grows as $\Theta(n^4)$ (dominated by W_{xh}: hidden $\times$ input $= O(n) \times O(n^2)$). This suggests that for large n, a sparse or convolutional architecture would be needed—but the Petri net representation itself scales cleanly.

Board	Places	Transitions	Hidden	Parameters
3×3	29	18	33	2,691
4×4	50	32	45	5,792
5×5	77	50	56	10,354
6×6	110	72	67	16,822

Table 1: NxN TTT Petri net and RNN scaling. Places $= 3n^2 + 2$, transitions $= 2n^2$, hidden $= \lfloor 8\sqrt{2n^2} \rfloor$.

Conjecture 14 (Observer–Readout Correspondence)**.** *The core-observer decomposition of a Petri net and the hidden–output decomposition of an RNN trained on that net are structurally analogous: in both cases, the readout layer (observer transitions or W_{hy}) is a projection over state that is already complete in the representation it reads from (core places or hidden state $\mathbf{h}_t$).*

In TTT, the evidence is suggestive:

1. *The observer's win-detection transitions consume cell places (`x00`, `x01`, ...) that were produced entirely by core play transitions. The observer adds no information—it pattern-matches on state the core already assembled.*

2. *The RNN's hidden state, when used directly as logits (Section 8.4), produces coherent play without the output projection W_{hy}. The hidden representation is already organized around the transition space.*

If the correspondence holds generally, it would imply that for any Petri net with a core-observer decomposition, an RNN trained on the core's marking dynamics will develop internal representations aligned with the core's transition space, and the observer layer's function (classification, termination detection, constraint checking) will be implicitly encoded in the hidden state before any explicit readout.

This remains unproven. The two systems could exhibit similar redundancy for different reasons: the Petri net's is structural (the incidence matrix determines information flow), while the RNN's is learned (gradient descent finds it). A proof would need to show that the RNN's loss landscape, shaped by the net's firing semantics, necessarily induces hidden-state alignment with the transition space.

9 Application: The Category Settle

The core-observer decomposition finds its most natural application in payment settlement networks, where the tropical eigenvalue answers the central design question: *what is the maximum sustainable transaction rate?*

9.1 The Category Settle

We define **Settle** as the free symmetric monoidal category generated by a payment Petri net.

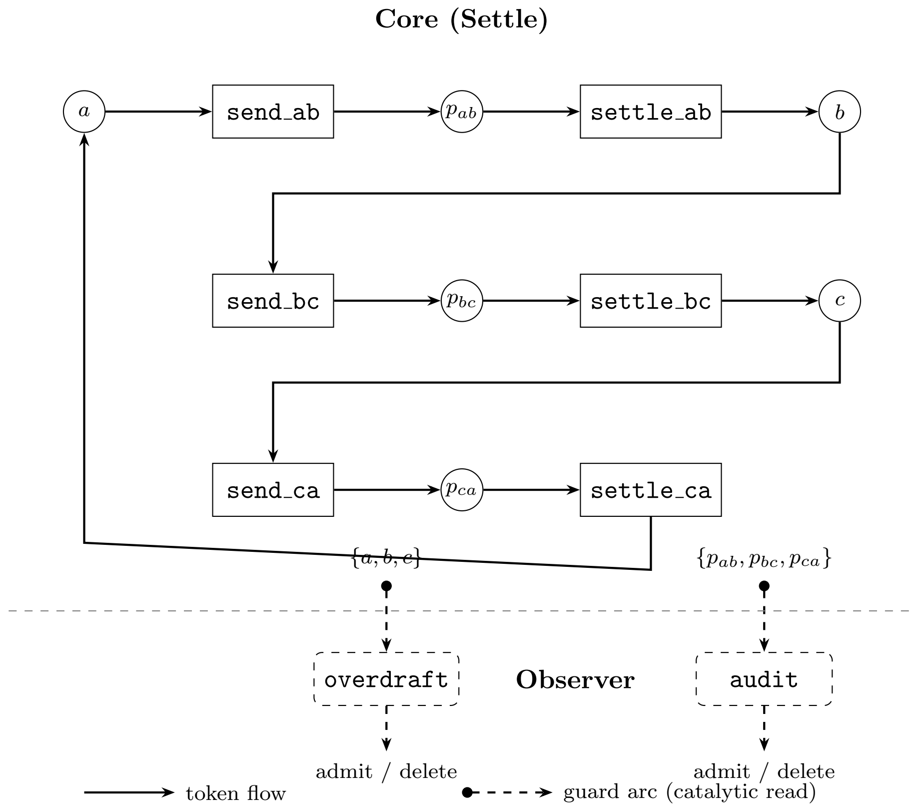
Definition 15 (The category **Settle**)**.** **Settle** *is the free symmetric monoidal category with:*

- ***Objects:*** *multisets of account places (token types representing balances and pending amounts).*
- ***Morphisms:*** *transitions (send, settle) generated by the incidence matrix C.*
- ***Composition:*** *sequential firing (*`send` ; `settle` = `transfer`*).*
- ***Tensor product:*** *parallel firing (concurrent independent settlements).*
- ***Symmetry:*** *token fungibility (one dollar is interchangeable with another).*

The monoidal structure enforces conservation: for every core morphism, the total weight of input wires equals the total weight of output wires. Double-entry bookkeeping is a theorem of **Settle***, not a policy imposed on it.*

9.2 String Diagram as Protocol Specification

The following string diagram declares a three-party circular settlement network. Each wire is a token type (account balance or pending payment); each box is a transition (send or settle). The diagram *is* the protocol specification—the incidence matrix C is its compilation.



Solid wires and boxes are the core category **Settle**; dashed elements are the observer. The core is a timed event graph: every place (wire) has exactly one producer (box output) and one consumer (box input). Dashed arcs are *guard arcs* (catalytic): the observer reads balance and pending places without consuming tokens. The observer determines whether each candidate morphism is admitted to **Settle** or deleted.

9.3 Incidence Matrix and Throughput

The core subnet has incidence matrix $C_{\text{core}} \in \mathbb{Z}^{6\times 6}$:

$$C_{\text{core}} = \begin{array}{c} \\ \texttt{send\_ab} \\ \texttt{settle\_ab} \\ \texttt{send\_bc} \\ \texttt{settle\_bc} \\ \texttt{send\_ca} \\ \texttt{settle\_ca} \end{array} \begin{array}{cccccc} a & b & c & p_{ab} & p_{bc} & p_{ca} \end{array} \atop \begin{pmatrix} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & -1 \end{pmatrix}$$

where a, b, c abbreviate `alice_bal`, `bob_bal`, `carol_bal` and p_{ab}, p_{bc}, p_{ca} abbreviate the pending places.

Theorem 16 (Settlement Throughput)**.** *The tropical adjacency matrix of C_{core} has eigenvalue $\lambda = 1$. The maximum sustainable settlement rate is one complete cycle (send ; settle) per unit time per channel.*

Proof. The transition dependency graph has a single Hamiltonian circuit:

$$\texttt{send\_ab} \to \texttt{settle\_ab} \to \texttt{send\_bc} \to \texttt{settle\_bc} \to \texttt{send\_ca} \to \texttt{settle\_ca} \to \texttt{send\_ab}$$

of length 6 and total weight 6 (each arc has weight 1). The maximum circuit mean is $\lambda = 6/6 = 1$. Every sub-circuit (e.g., `send_ab` $\to$ `settle_ab` $\to$ `send_bc` $\to$ `settle_bc` $\to$ `send_ca` $\to$ `settle_ca`) has the same mean. By Karp's theorem, the tropical eigenvalue is $\lambda = 1$. □

Remark 17. *The eigenvalue $\lambda = 1$ means the system is* balanced*: no channel is a bottleneck. In a heterogeneous network (different processing times per channel), λ would identify the slowest channel—the settlement bottleneck—without simulation. Changing the network topology (adding a fourth party, splitting a channel into two hops) changes λ, and the critical circuit tells the designer exactly which path constrains throughput.*

9.4 Conservation as Accounting

The kernel of C_{core} yields the P-invariant:

$$a + b + c + p_{ab} + p_{bc} + p_{ca} = \text{const}$$

This is the *accounting identity*: the total money in the system is conserved across all firings. No audit is required to verify this—it is a structural theorem of the incidence matrix. Every token debited from one account is credited to a pending place, and every token cleared from pending is credited to the recipient.

In a traditional payment system, this invariant is enforced by policy (double-entry bookkeeping procedures) and verified by audit. In **Settle**, it is enforced by the monoidal structure: a morphism in **Settle** *cannot* create or destroy tokens, because the generators (transitions) are defined by C, and C has the invariant in its kernel.

9.5 Observer as Existence Gate

The observer does not audit transactions after the fact. It determines whether a payment object *should exist at all.*

A candidate payment is a morphism in the ambient category of structurally valid transfers—any transition the incidence matrix permits. The observer reads the core net's state via guard arcs (catalytic: they test a place's marking without consuming tokens) and applies business rules. If the guard is satisfied (e.g., balance $\geq$ amount), the morphism is admitted to **Settle**. If not, the candidate is *deleted*—it never becomes a morphism, never composes, never appears in the event log.

Observer	Rule	Effect
`overdraft`	$\texttt{bal}_{\text{src}} \geq \texttt{amount}$	admit or delete
`audit`	pending $\leq$ limit	admit or delete

Categorically, the observer is a *sieve* on the ambient category: a subfunctor of the representable presheaf that selects only the morphisms satisfying the business predicate. **Settle** is the image of this sieve—the subcategory of payments that passed. Invalid payments are not rolled back, reversed, or compensated. They do not exist in **Settle** because they were never admitted.

This is the payment-network analogue of the TTT observer, with one structural difference:

	TTT Observer	Payment Observer
Reads	cell history places	account balance places
Mechanism	consuming (game ends)	catalytic / guard (system continues)
Timing	post-hoc (after move)	pre-hoc (before commit)
On failure	game continues	morphism deleted

The TTT observer consumes cell tokens because the game terminates—win detection is a terminal observation. The payment observer reads catalytically because the system must continue after each transaction. The guard arc is the formal mechanism: it appears in the incidence matrix as a row that tests but does not decrement, gating the transition without altering the marking.

9.6 ZK Privacy

The incidence matrix C compiles directly into an R1CS circuit (Section 2). A Groth16 proof of a settlement transition proves:

> *"A valid transfer of x tokens occurred from account A to account B, and both accounts had sufficient balance."*

without revealing x, A's balance, or B's balance.

The prover reveals only the pre- and post-state Poseidon hashes; the verifier checks the 128-byte proof in constant time. This is the payment-network analogue of the TTT ZK proof (Section 2): the same circuit structure, the same incidence matrix encoding, the same verification cost—applied to a domain where privacy is a legal requirement, not a game mechanic.

10 Related Work

Tropical analysis of Petri nets. The connection between timed event graphs and max-plus algebra is classical [1, 7]. Our contribution is the decomposition into tropical (core) and non-tropical (observer) subnets, and the use of this decomposition to characterize which parts of a system admit spectral analysis.

Categorical semantics of Petri nets. Meseguer and Montanari [10] proved that the firing semantics of a Petri net is a free symmetric monoidal category, with transitions as generators and token swaps as the symmetry. Sassone [13] refined the construction. Baez and Master [2] extended the framework to *open* Petri nets—composable systems glued along shared boundary places—which is directly relevant to payment channel networks where settlement layers compose. The category **Settle** (Section 9) is a specific instance of this construction, where the monoidal conservation law is the accounting identity.

Petri net decomposition. Subnet decomposition of Petri nets has a long history [12], primarily for analysis tractability. The core-observer decomposition is distinguished by the sink property, which gives it a categorical interpretation as a functor.

ZK proofs for state machines. ZK proofs for state-machine transitions have been explored in blockchain contexts [3]. The Petri net formulation gives a uniform treatment: the incidence matrix is both the state-machine specification and the R1CS template.

Kolmogorov complexity and structure. The connection between Kolmogorov complexity and structural description has been explored in algorithmic information theory [11]. Our "earned compression" formalization makes this connection concrete for a specific class of discrete-event systems.

11 Discussion

Convergence as proof. The strongest result in this paper is Theorem 1: three formalisms independently discover the same boundary. A skeptic could dismiss any single formalism's decomposition as an artifact of its assumptions. But when ODE simulation (continuous, mass-action kinetics), tropical algebra (discrete, max-plus semiring), and zero-knowledge proofs (cryptographic, R1CS constraints) all identify the same partition of transitions, the partition is not an artifact—it is the structure.

This is analogous to how multiple independent measurements of the same physical constant provide stronger evidence than any single measurement. The formalisms share no mathematical machinery: the ODE uses real-valued differential equations, the tropical analysis uses the max-plus semiring, and the ZK system uses finite-field arithmetic. Their convergence is the proof.

The incidence matrix as universal interface. The practical consequence is that C is a *universal interface*. A single 35×33 integer matrix encodes the full TTT game: its legal moves (firing rules), its strategic values (ODE equilibrium), its throughput (tropical eigenvalue), its conservation laws (null space), and its transition proofs (ZK circuit). No other representation achieves all five with the same data.

Irreducible complexity. The observer's non-tropical structure is not a failure of analysis but a *feature of the domain*. Win detection in tic-tac-toe is irreducibly non-tropical because a single cell participates in multiple win lines. The boundary between compressible and incompressible structure is itself informative—it tells you where the domain's intrinsic complexity lives.

From games to finance. The TTT example demonstrates the decomposition on a system where throughput is irrelevant (a game has exactly 9 moves). The **Settle** example (Section 9) demonstrates it on a system where throughput *is the product*: a payment network's value is directly proportional to its settlement rate. The same incidence matrix formalism serves both, but the tropical eigenvalue transforms from a structural curiosity (TTT: $\lambda = 1$ confirms alternation) to a design parameter (payments: λ determines capacity).

The core-observer decomposition also gains practical meaning: the core is the settlement engine (move money), the observer is the existence gate (determine whether a payment should exist at all). A candidate transfer that would overdraft an account is not rejected, rolled back, or compensated—it is *deleted*: it never becomes a morphism in **Settle**. In traditional finance, settlement and compliance are separate systems maintained by separate teams with separate audits. In **Settle**, they are a categorical decomposition of a single incidence matrix—separation is a theorem, not an organizational choice, and the observer's verdict is existential, not remedial.

Generalization. The core-observer pattern appears in any system with a computational core and a monitoring/termination layer:

- **Payments:** core = settlement, observer = compliance/audit (Section 9).
- **Games:** core = move execution, observer = win/loss/draw detection.
- **Workflows:** core = task execution, observer = timeout/error detection.
- **Protocols:** core = message passing, observer = violation/completion checks.

In each case, the core admits tropical analysis (throughput, cycle time) and the observer admits only discrete analysis (reachability, enabledness). The formalism-invariance theorem predicts that this same boundary will be discovered independently by any sufficiently expressive analysis framework applied to these domains.

12 Conclusion

The core-observer decomposition of Petri nets is invariant across ODE simulation, tropical analysis, and zero-knowledge proof. Three formalisms with no shared mathematical machinery independently discover the same structural boundary: a partition into a compressible core (timed event graph, stable ODE, uniform R1CS) and an incompressible observer (multi-consumer places, sink dynamics, non-uniform witnesses). The convergence of three independent discoveries is stronger evidence than any single proof could provide.

The incidence matrix C is the earned compression of the domain: the shortest description from which all behavioral properties can be recovered. Each recovery is proved exact, and the tightness of each compression stage follows from classical results in tropical algebra, chemical reaction network theory, and succinct argument systems. The category **Settle** demonstrates that this is not merely theoretical: the string diagram of a payment network is simultaneously its protocol specification, its throughput certificate (via λ), its accounting proof (via $\ker C$), and its ZK circuit template. The boundary between what compresses and what doesn't is itself the most informative structural feature of the system.

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